

University *of* Victoria Department *of* Mechanical Engineering

QT1.H Fuel System – Final Report
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Prepared For:

MECH 400b: Capstone Design Project

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1.0 Introduction

University of Victoria's Centre for Aerospace Research (CfAR); the client, requested the design and build of a fuel system for their QT1.H hybrid research uncrewed aerial vehicle (UAV), to improve on the current off-the-shelf solution. Improvements were to be made in slosh mitigation, fuel pickup delivery and venting, and heat transfer, all while mitigating added mass and remaining within the constrained available volume. A custom solution was proposed and built, using two thermo-formed high-density polyethylene (HDPE) tank shells, joined by an aluminum mounting structure. To address the project goals, open cell foam was used for slosh mitigation, custom low-profile fuel fittings were used to maximize available volume, and a heat shield was used to mitigate radiative heating from the nearby engine. The scope of the project was limited to the fuel system of the aircraft, including the fuel tank, fuel connections, and accommodations for the existing fuel pump.

2.0 Design Description

The delivered fuel system can be broken down into four main subsystems: the tank shell and structure, slosh mitigation mechanisms, the fuel delivery system, and heat transfer mitigation. The respective designs and functionalities of these systems will be explored in the following subsections.

2.1 Tank Shell and Structure

During the design process, the first challenge was to identify the material that would comprise the tank shell; this would drive the design of remaining components. Several options for tank material were explored including Aluminium, composite materials, and plastics. Ultimately HDPE was selected, as it adequately met all requirements and constraints, without presenting issues such as the cost associated with Aluminium welding or the compatibility issues often experienced between gasoline and composite materials.

The tank shell was initially designed such that it could be manufactured using either thermoformed plastic or composite material, such that if any challenge were encountered during manufacturing, the material could be changed without change to the design of any other components. This drove the final tank design wherein the shell is split into two parts. These two halves are adjoined by an Aluminium structure which acts as a backbone for mounting and seals the shell together using O-rings.

To contain fuel in a weight efficient manner, while allowing for continued internal access to the fuel tank, two HDPE shells are united by a central Aluminium structure. These shells are thermoformed, using a CNC milled wooden mold. The nominal thickness of the HDPE prior to thermoforming was $1/16[in]$; no excessive change to material thickness was observed prior to manufacturing. The two halves of the tank shell are split on the horizontal plane. The bottom portion of the tank shell possesses a ten (10) degree angle towards the rear of the tank, as to accommodate for any pitch angle encountered during flight.

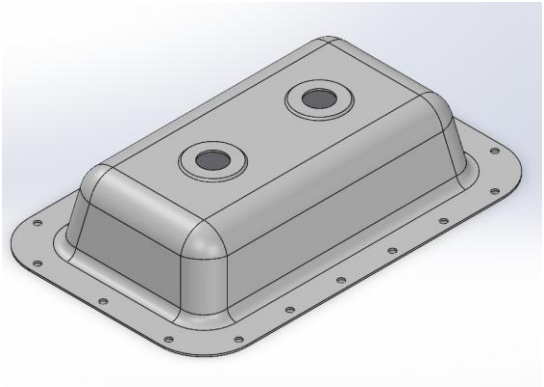


Figure 1: Top of Tank Shell

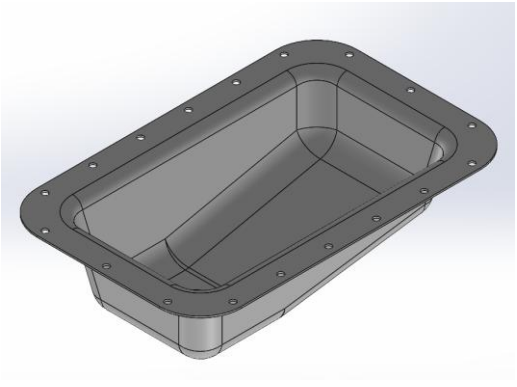


Figure 2: Bottom of Tank Shell

An Aluminium structure seals these two halves together. This structure is comprised of three pieces: two clamps and the central girder. The central girder provides the structure for mounting of the tank within the aircraft fuselage. Meanwhile, the clamps distribute clamping force, holding the two parts of the shell together. The assembly of the two parts of the tank and the Aluminium structure can be seen in Figure 3.



Figure 3: Fuel Tank Shell & Structure

The central Aluminium girder is manufactured using 2.5-D CNC milling, from quarter-inch nominal Aluminium stock. It measures approximately five inches in width by seven inches in length. It includes an O-ring groove on each side, for sealing, and an array of clearance holes, both for mounting to the fuselage and sealing. The clamping structures are manufactured from one sixteenth-inch nominal Aluminium. They possess the same overall dimensions as the central structure, with far more simple geometry. The clamping structures feature only clearance holes for sealing. The central structure, and clamping structure are shown in Figure 4, and Figure 5 respectively.

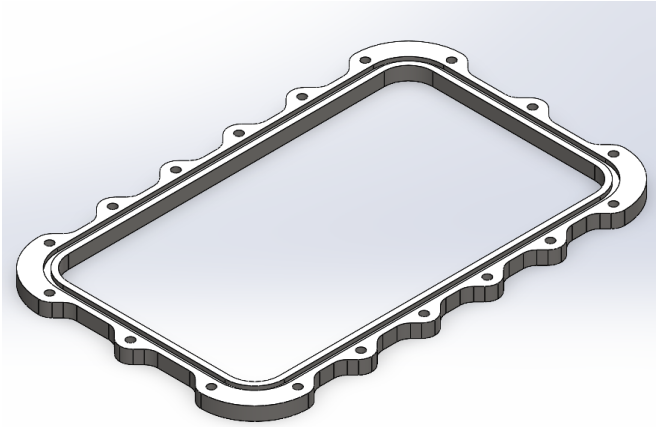


Figure 4: Central Aluminum Structure

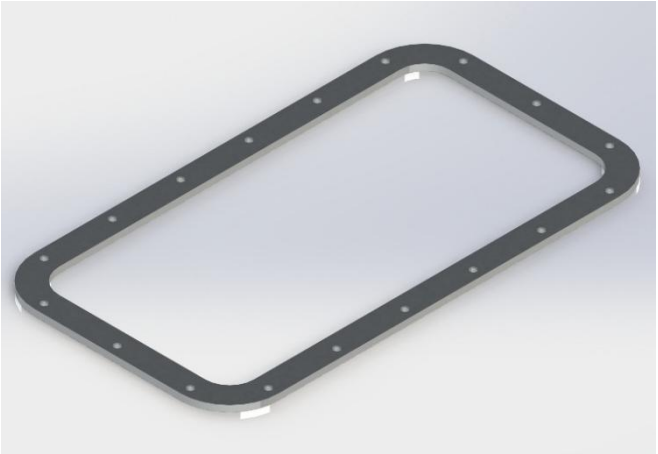


Figure 5: Aluminium Clamping Structure

2.2 Slosh Mitigation

During the design and design review process, numerous mechanisms for slosh mitigation were explored, including baffles of varying configuration, open cell foam, and combinations thereof. It was identified by CfAR that the baffle design exhibited an excessively high safety factor relative to requirements and consumed a significant portion of internal tank volume, while adding unnecessary mass. This led to the selection of fuel-rated open-cell foam as the main means of fuel slosh mitigation. Shown in Table 1 are the design objectives associated with fuel slosh mitigation

that led the client to advise the use of foam, details and rationale for item in Table 1 are included in the attached objectives document.

Table 1: Design Objectives Pertinent to Slosh Damping Design

ID	Description	Metric	Target Value	Level of Importance
1.	Minimize Added Mass	Maximum fuel system mass (including fuel):	5 $[kg]$	Required
2.	Minimize Added Mass	Target fuel system mass (including fuel):	2.5 $[kg]$	Ideal
3.	Thermally Insulate Tank and Fuel	Maximum temperature:	50 $[°C]$	Required
4.	Limit Fuel Sloshing	Maximum time for half of fuel volume to traverse from one extreme position to the other	1 $[s]$	Required

The mass and slosh related design objectives established by CfAR at the outset of the project are listed above in order of relative importance. One key objective states that aircraft exceeding 25 $[kg]$ maximum take-off weight (MTOW) are subject to more restrictive regulations. Maintaining a system mass below 2.5 $[kg]$ was identified as the ideal target to keep the aircraft under this threshold.

Throughout the design process, the fuel system remained below the 2.5 $[kg]$ target, which initially supported the slosh baffle concept. However, CfAR clarified that the tank should remain as light as reasonably achievable, and that any added mass must be explicitly justified regardless of whether the system remains under the ideal threshold. In this context, it was determined that the performance benefits of the baffle system could not be clearly demonstrated. Designing and validating a dedicated slosh test rig is outside the scope of this project, and therefore no evidence can be presented to show that baffles outperform foam. Furthermore, CfAR has extensive experience using fuel-rated foam in other high-performance applications, including jet-propelled aircraft that experience significant fuel excitation due to high levels of thrust. This led to the selection and implementation of the open-cell foam slosh mitigation design.

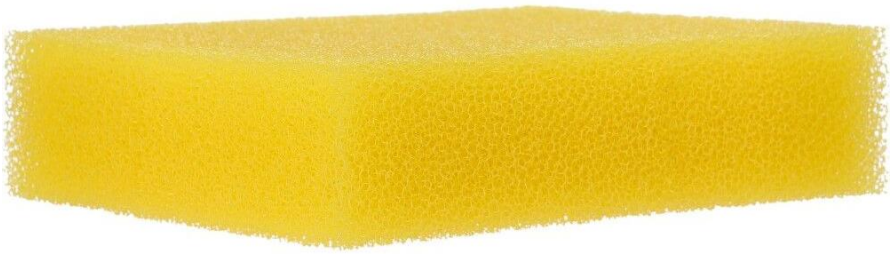


Figure 6: Type II Slosh Damping Foam

Shown in Figure 6 is the foam implemented in the final design. This material is classified as Type II slosh damping foam and consists of an open-cell, reticulated structure that allows fuel to freely permeate while disrupting bulk fluid motion. The foam occupies less than 3% of the tank volume [1], thereby minimizing loss of internal tank volume. The material is manufactured from fuel-compatible polyurethane and is designed specifically for continuous exposure to hydrocarbon fuels.

2.3 Fuel Delivery

The fuel system is responsible for the management of fuel, air, and vapour flow during flight and filling. To achieve reliable fuel pickup during all flight regimes, the selected design includes a single fitting at the bottom-most point at the rear of the tank. This configuration ensures fuel flows to the desired pickup location, irrespective of pitch and roll, when within the expected flight conditions.

Fill and vent lines connect exterior bulkhead fittings on the top of the tank to two fittings on the shell of the fuselage. The fill line allows fuel flow from the external fill port into the tank. The ground side fueling system consists of a fuel can, fuel pump, and commercial quick disconnect fitting, which are specified and exterior to the scope of this project. The vent line allows excess air to leave the tank during fueling and maintains atmospheric pressure inside the tank as exterior pressure changes, temperature changes, or fuel is consumed. The vent line is connected from the top of the tank to the bottom of the fuselage, ensuring that if the tank is overfilled excess fuel will spill outside the vehicle.

Commercially available fittings were largely incompatible with the geometric constraints of the project or were incompatible with gasoline. As such, custom low-profile fittings were designed and manufactured. These custom fittings allowed fuel lines to be directed tangentially to the surface of the fuel tank, without requiring the protrusion of a barbed bulkhead fitting, or the bend radius of the fuel hose. This is all achieved within one (1) inch of the tank surface. The custom fitting is shown below in Figure 7 and Figure 8.

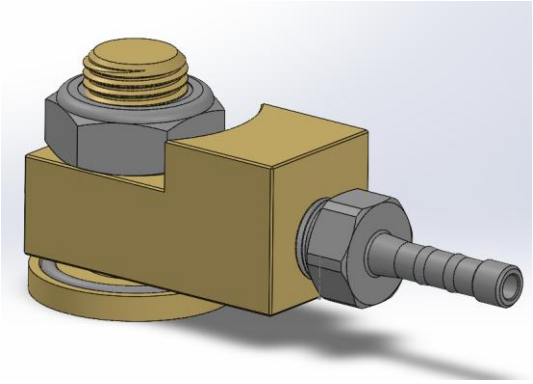


Figure 7: Custom Fitting Exterior

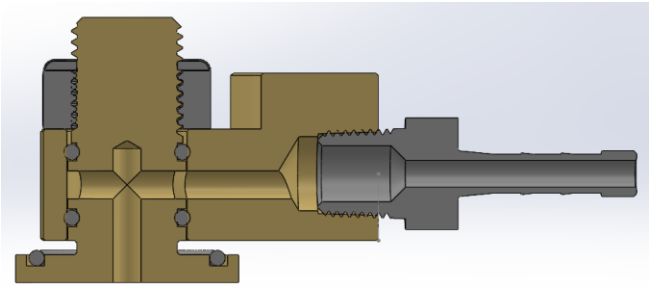


Figure 8: Custom Fitting Section View

The general configuration of the fuel system remains unchanged, and is shown below, in Figure 9.

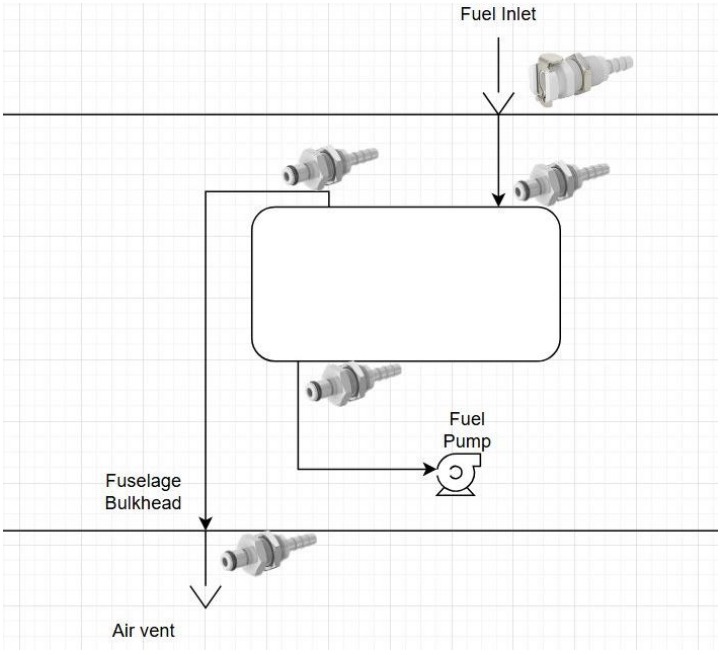


Figure 9: Fuel System Piping Diagram

2.4 Mounting

The fuel tank mounting system within the aircraft fuselage must support both the fuel tank and the fuel pump while accommodating the existing components, including the air inlet duct, coolant pump, and radiator. The mounting hole locations are fixed, as the aircraft's base plate has already been manufactured with pre-drilled mounting holes.

The mounting systems uses a post-and-plate geometry consisting of four aluminum posts that connect the base plate to a middle plate. The middle plate, along with a dedicated mounting bracket, serves as the mounting surface for the fuel pump and provides structural stability. The heat shield will be secured to the middle plate as well. Four additional aluminum posts extend from the middle plate to the clamped portion of the fuel tank.

A post-and-plate mounting structure was chosen due to its proven structural integrity and ease of assembly. Aluminum was chosen because of its high strength-to-weight ratio and ease of manufacturability. Given that mass is a key design constraint, aluminum provides a significant weight advantage compared to steel. Nominal material sizes were selected to simplify manufacturing, with 3/8-inch diameter posts and ¼-inch thick plates. The fasteners are 6-32 hardware, consistent with the fuel tank fastener choice.

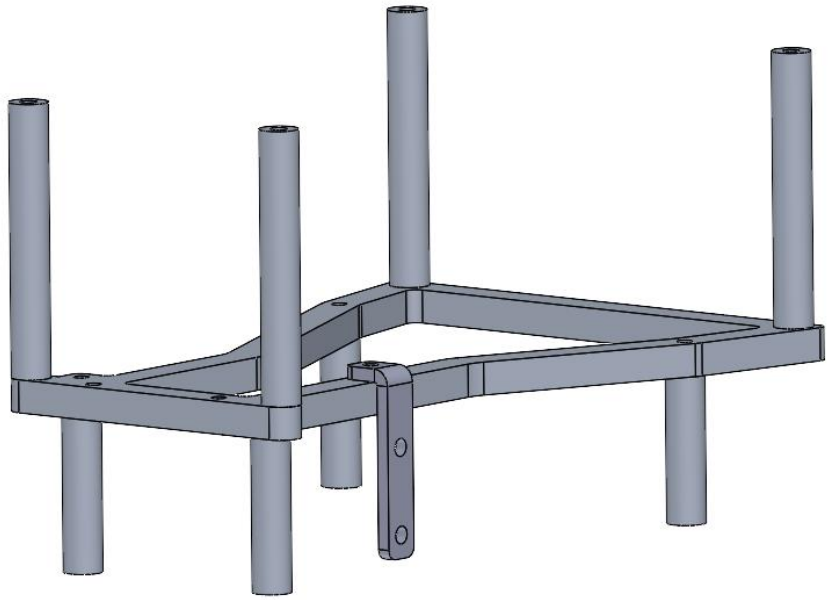


Figure 10: Mounting Assembly

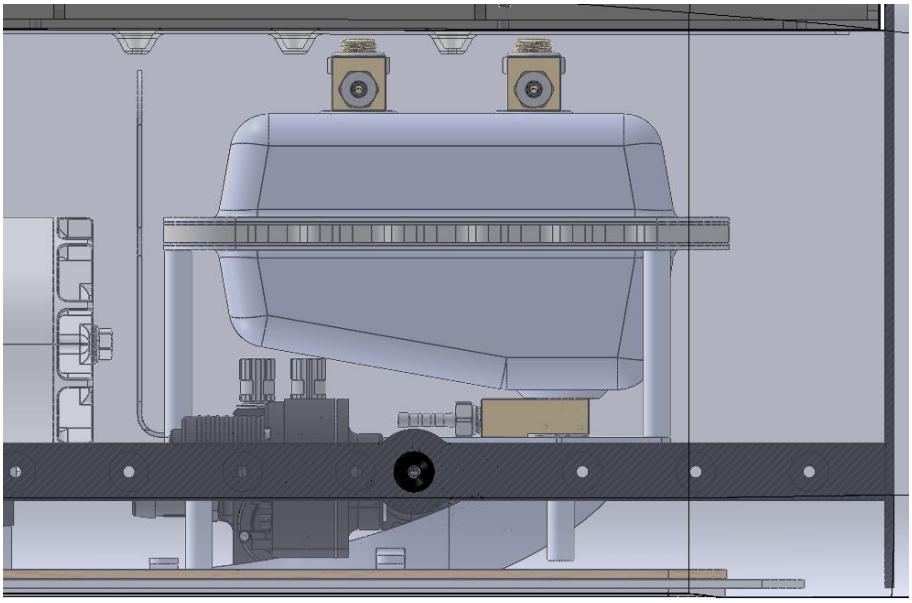


Figure 11: Mounting Assembly Inside QT1.H Airframe

While the overall profile and design of the mounting system remain largely consistent with the previous revision, several modifications were made following the most recent design review with CfAR. It was identified that the mounting assembly would need to accommodate the coolant pump unit (CPU), radiator, and a 20mm air gap to allow airflow through to the radiator. These additional constraints caused changes to the overall dimensions and profile of the assembly, with the middle plate redesigned to clear the CPU and the posts configured to provide access to electrical connections for all existing components while preventing interference with surrounding hardware.

Two further design changes were introduced to improve manufacturability and thermal performance. First, a dedicated aluminum bracket was incorporated to constrain the fuel pump, Figure 12, replacing the original design which required cross-drilled holes in the middle plate. Given the fuel pump's small size and low mass, the bracket is secured using a single 6-32 bolt and nut, which provides sufficient securement. Second, the heat shield mounting was relocated from the aluminum clamping structure of the fuel tank to the middle plate, minimizing direct heat conduction into the fuel tank.

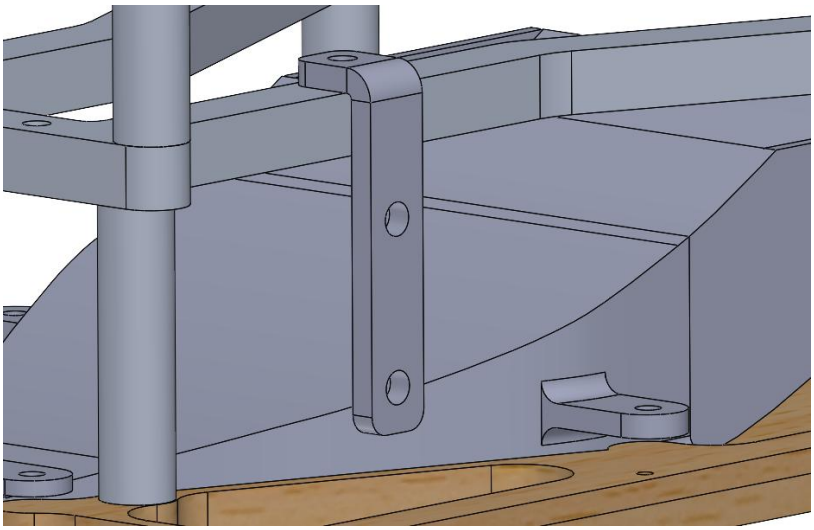


Figure 12: Fuel Pump Mounting Bracket

2.5 Heat Transfer Mitigation

One of the main challenges with the design of this fuel system was its proximity to the aircraft engine and generator. As a result, heat deflection was required to protect the fuel tank from extreme thermal conditions. Once analysis was performed to calculate the heat transfer rate (as described in the section below), it was clear a heat shield was necessary.

Several different heat shield designs and materials were considered. While materials such as Inconel, titanium, and other superalloys were considered due to their superior strength-to-weight ratios, their material costs were unjustifiable for this design. Aluminum 5052 was selected due to its low cost, light weight, and suitable thermal properties, as documented in the analysis section. While the radiative heat transfer rate from the engine is high, it does not warrant temperatures where aluminum would be subject to deformation. Modal studies were run to determine an appropriate thickness value. After iterating through available nominal thicknesses from McMaster-Carr, it was concluded that 0.063" (1/16 of an inch) nominal thickness was optimal for balancing weight with vibrational amplitude risks.

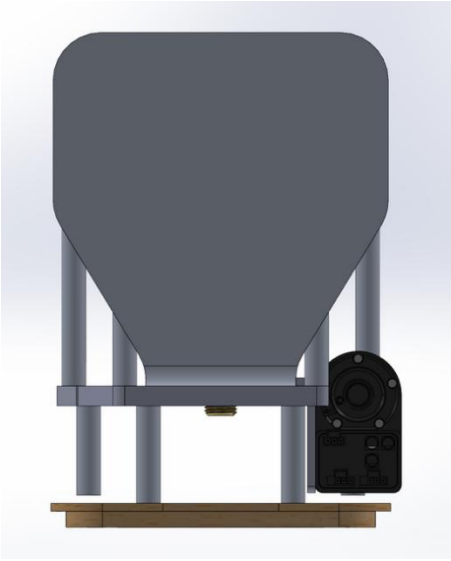


Figure 13: Front Line of Sight Tank Blockage

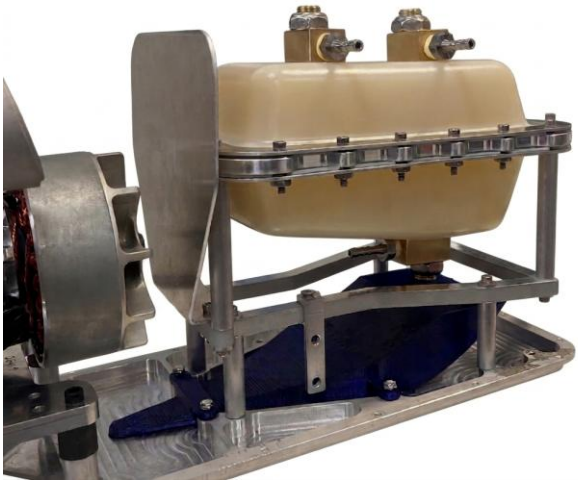


Figure 14: Fuel Tank with Heat Shield

The final heat shield design is shown above. This design was selected as it blocks all line of sight from the engine. The lower half was cut to match the tapering nature of the fuel tank as needed by HDPE drafting requirements. The lower half was cut to match the tapering nature of the fuel tank as required by HDPE drafting standards. The upper half of the engine contains exhaust manifolds which extend laterally in the upper half of the fuselage. As a result, tapering the upper half of the heat shield was not possible if it was to prevent line of sight.

Several alternative designs for heat rejection were considered. While the geometric tapering profile of the heat shield changed as the tank design evolved, the final decision was to extend the shield without tapering on the top half. Several ultralight options were also presented to CfAR, but the final decision warranted erring on the sight of caution. This ensures all line of sight on the upper portion is blocked, as exhaust manifolds are one of the highest temperature components of an engine. Ultimately, the selected Aluminum 5052 design provides a cost-effective and structurally sound solution for thermal protection. By prioritizing a full line-of-sight blockage of the exhaust manifolds and optimizing material thickness through modal analysis, the final heat shield successfully balances weight reduction with the critical safety requirements of the fuel system.

3.0 Analysis

The following section details the structural, thermal, and vibrational analyses completed to validate the fuel system design. The SolidWorks simulation environment was utilized to assess the mounting assembly, clamp plate, heat shield, and tank shells under operational loading conditions, ensuring all components meet the required factors of safety and thermal performance targets.

3.1 Structural Analysis

The following structural analyses were conducted to validate the clamp plate and mounting assembly under their respective loading conditions. SolidWorks FEA was used to evaluate stress distributions, deflections, and factors of safety, confirming that both components perform as intended under operational loads.

3.1.1 Clamp Plate

A structural FEA analysis was conducted on the clamp plate to confirm that O-ring compression forces would not cause plate deformation, gapping, or permanent yielding. Bolt strength calculations first verified that the 20 retention bolts provide sufficient clamping force, yielding a factor of safety of 45 against yield.

Table 2: Retention Bolt Strength Calculations

O-Ring Stiffness		
Parameter	**Value**	**Units**
Thickness	2.62	$[mm]$
Squeeze	20	$[\%]$
Force/Length	1.7	$[N/mm]$
OD	195.32	$[mm]$
ID	190.24	$[mm]$
Dm	192.79	$[mm]$
Total Force	1029.6	$[N]$
#6-32 Tensile Area	5.81×10^{-6}	$[m^2]$
Number of Bolts	20	$[-]$
Stress	8.87	$[MPa]$
Yield Strength	400	$[MPa]$
Factor of Safety	45.1	$[-]$

Full details of the simulation setup, boundary conditions, and load application are documented in the Detailed Design Documentation report.

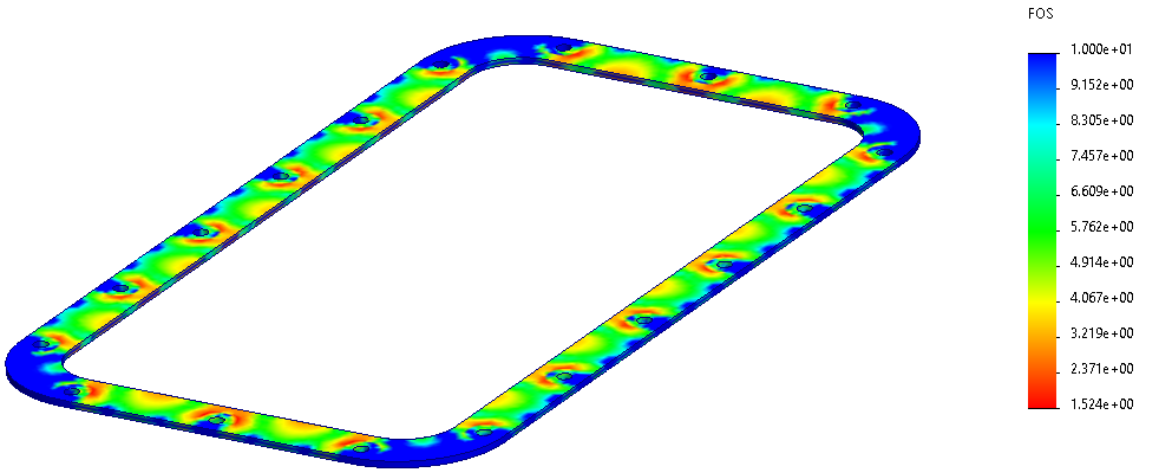


Figure 15: Clamp Plate Factor of Safety Results

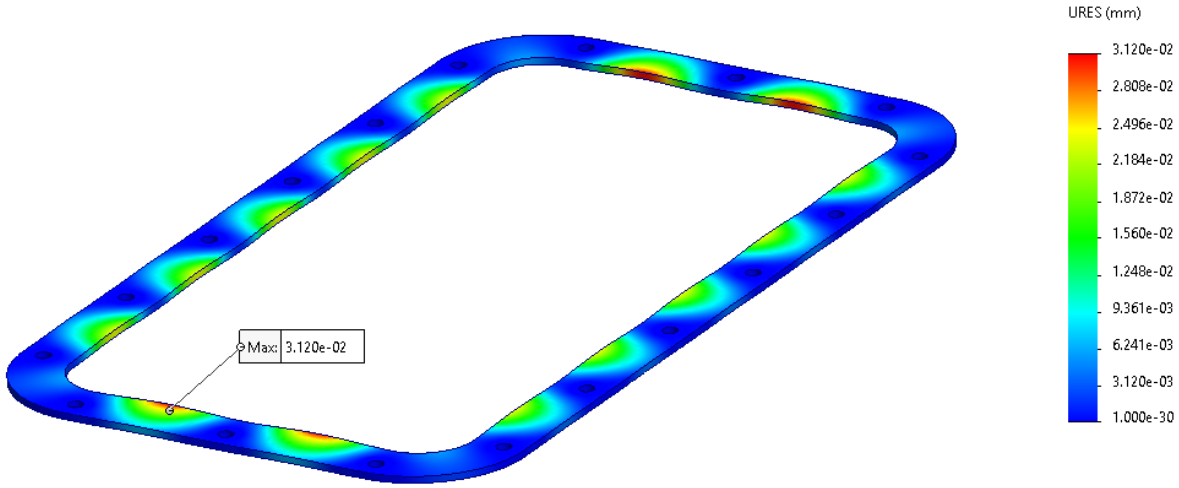


Figure 16: Clamp Plate Deflection Results

The FEA results confirmed that 1/16" aluminum is sufficiently stiff, with a maximum deflection of one thousandth of an inch and a minimum factor of safety of 2.5, both well within acceptable limits. These results validate the clamp plate design for use in the final prototype.

3.1.2 Mounting Assembly

A structural FEA analysis was conducted on the mounting assembly to confirm it could withstand the 9G pneumatic launch load case. Using Newton's second law, forces were calculated as 31.9 N and 39.4 N for the top and bottom posts respectively, accounting for the difference in supported mass between the two tiers. A fixed constraint was applied at the baseplate interface, with bonded component interactions used to simplify the simulation. Full details of simulation setup, boundary conditions, and load application are documented in the Detailed Design Documentation.

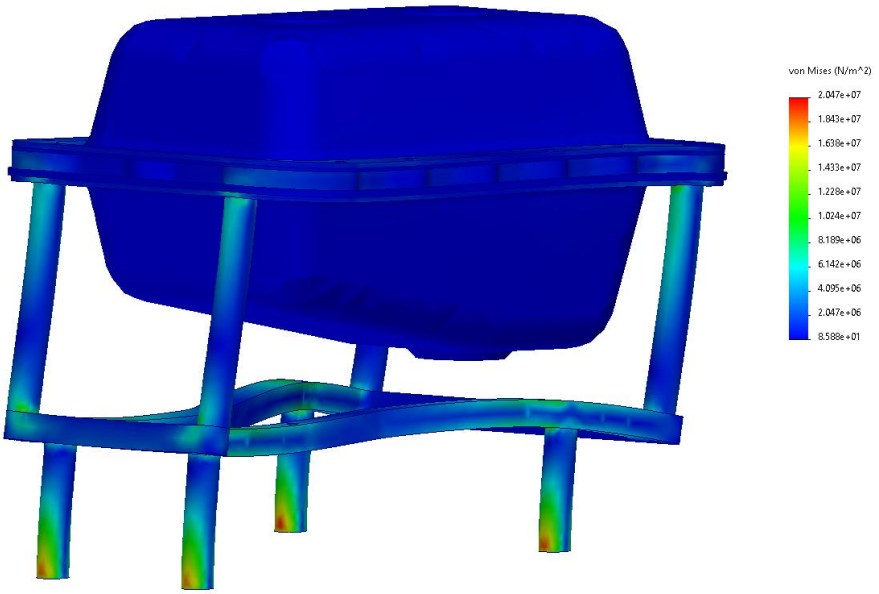


Figure 17: Stress Distribution

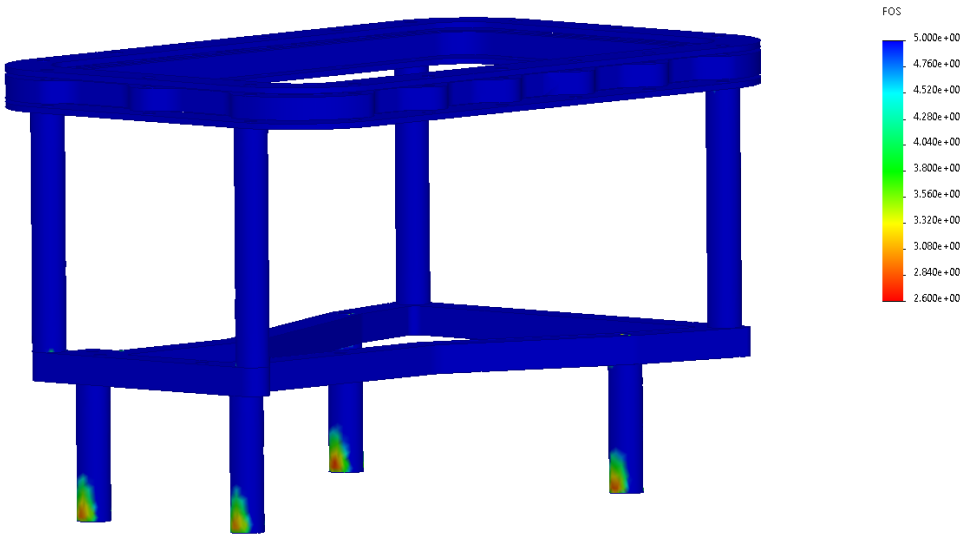


Figure 18: FOS Plot

The FEA results show a maximum Von Mises stress of 20.5 MPa at the base plate junction, yielding a minimum factor of safety of 2.6 against the 276 MPa yield strength of 6061 aluminum. These results confirm the mounting assembly performs as intended under operational loads and validate the design for use in the final prototype.

3.2 Heat Transfer Analysis

A steady-state thermal simulations was conducted in SolidWorks to evaluate the radiative heat risk posed by the engine exhaust manifold on the fuel tank. Using the Stefan-Boltzmann radiation equation with a cast iron emissivity of 0.75 [2] and a conservative view factor of 0.3, the radiative heat flux from the manifold operating at 400°C was calculated to be 2478 W/m^2. Heat flux was applied to tank surfaces directly facing the exhaust manifold, with a free convection coefficient of 10 W/m^{2}K applied across all exposed surfaces to simulate natural cooling. Two

simulations were conducted: one with and one without an aluminum heat shield. Full details of the simulation setup, boundary conditions, and load application are documented in the Detailed Design Documentation.

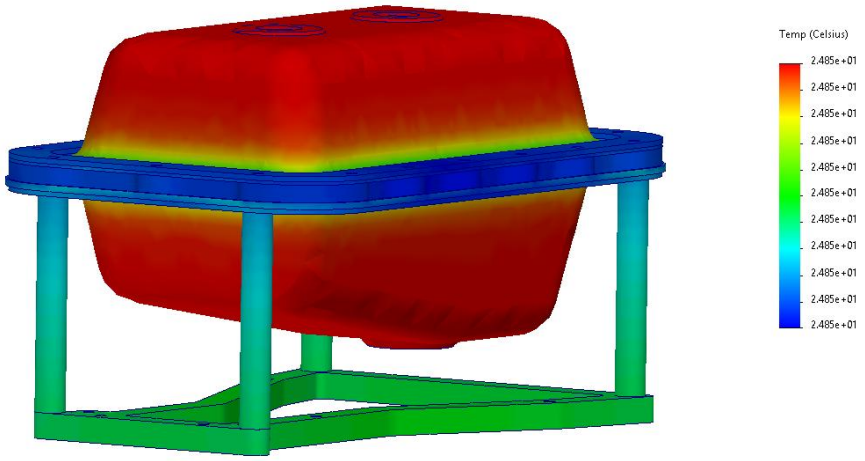


Figure 19: Thermal Simulation Results with No Heat Shield

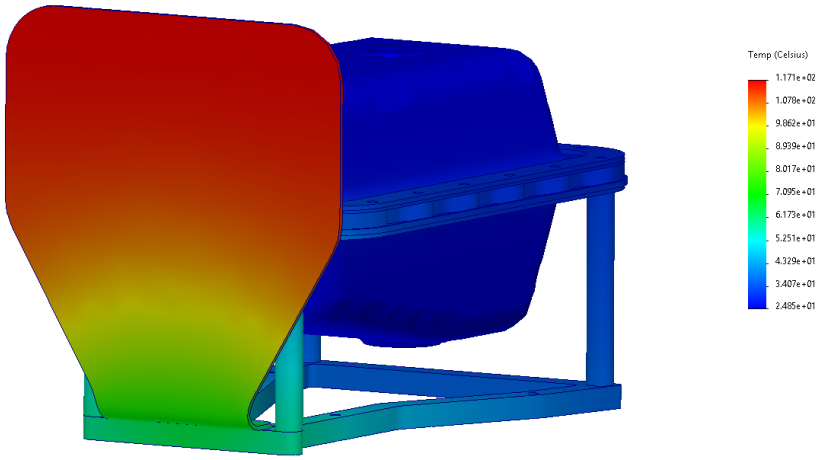


Figure 20: Thermal Simulation Results with Heat Shield

Without a heat shield, the simulation predicted a maximum HDPE temperature of 249°C, well above it 135°C melting point. HDPE's low thermal diffusivity further amplifies this risk, as it retains heat rather than dissipating it. With the aluminum heat shield included, the maximum tank temperature was reduced to 39°C, which is well within the safe operating temperatures for HDPE. The maximum temperature occurred at the clamped junction between the HDPE and aluminum (refer to Figure 21) due to conduction through the mounting assembly.

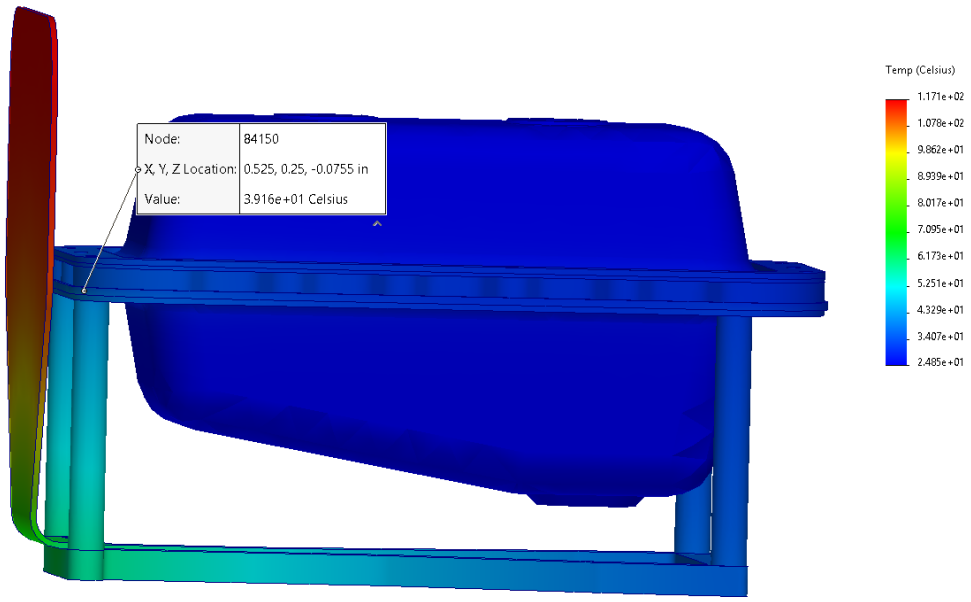


Figure 21: Location of Maximum HDPE Temperature

3.3 Modal Study

A modal analysis was conducted on the heat shield to assess the risk of resonance between its natural frequencies and the engine's harmonic excitation frequencies during cruise. The engine cruise speed of 5500-7000 RPM corresponds to excitation frequencies of 92-117 Hz for the basic harmonic, extending to approximately 700 Hz across six harmonics. The heat shield mounting flange was set as the fixed geometry boundary condition. Full details of simulation setup and boundary conditions are documented in the Detailed Design Documentation.

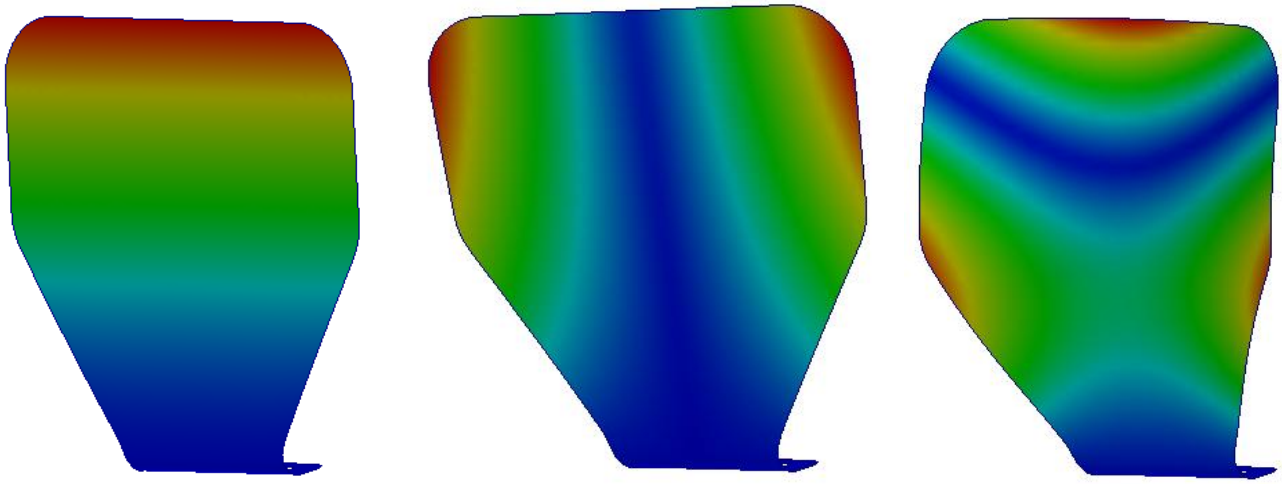


Figure 22: Modal Analysis Results (1)

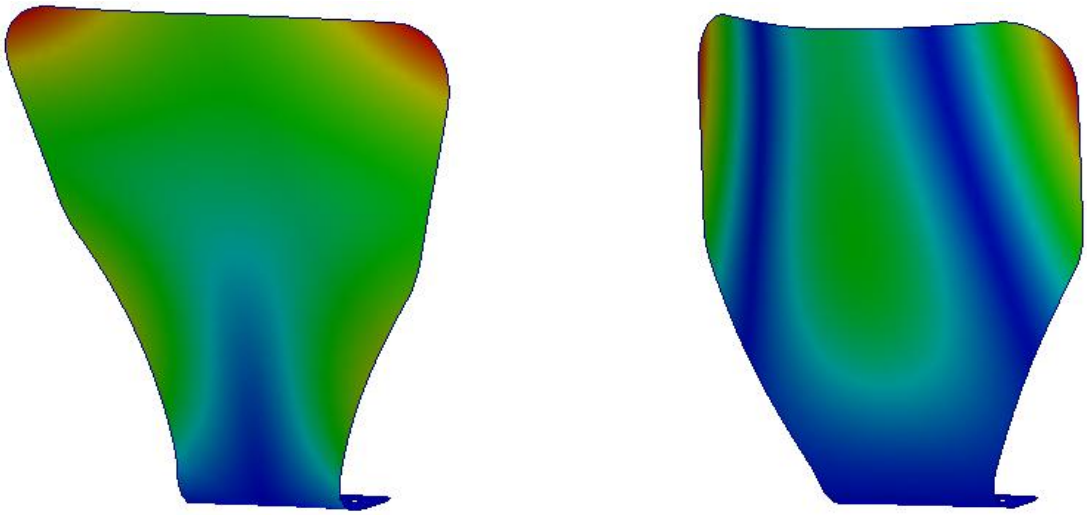


Figure 23: Modal Analysis Results (2)

Table 3: Engine Harmonics 1-6 & Frequencies

Engine Harmonic	Frequency Range [Hz]
1	92-117
2	184-233
3	276-350
4	368-467
5	460-583
6	552-700

Table 4: Heat Shield Modes 1-5 & Frequencies

Heat Shield Mode	Frequency Range [Hz]
1	60
2	142
3	416
4	542
5	611

The first five natural frequencies of the heat shield were found to be 60 Hz, 142 Hz, 416 Hz, 542 Hz, and 611 Hz. Modes 1 and 2 present low resonance risk under steady cruise conditions, with Mode 1 only a transient concern during engine start-up and shutdown. Modes 3, 4, and 5 directly overlap with engine harmonic bands 4, 5, and 6 respectively, representing resonance concerns under operation. Future experimental vibration testing is advised to validate these analytical predictions.

4.0 Risk, Reliability, and Safety Considerations

The following section outlines the risk, reliability, and safety considerations undertaken to identify and mitigate potential failure modes of the fuel system. A failure modes and effect analysis (FMEA) and hazard and risk

assessment (HARA) were conducted to evaluate failure causes, effects, and associated risk levels, with full details provided in the attached risk assessment document.

4.1 Failure Modes and Effect Analysis (FMEA)

To adequately evaluate possible failure modes of the fuel tank, a failure modes and effects analysis (FMEA) procedure, was undertaken. Subsystems were evaluated for possible failure mode, failure cause, failure effect, and presence of hazard. The failure modes were then assigned values for severity, occurrence, and detectability (whose scales are shown in Figure 24), resulting in their respective associated risk priority number (RPN) as well as the product of RPN and population. The failure modes were then assigned a repair action, recommended mitigation, and initial contingency cost. Due to changes in design, additional items were added to the FMEA table.

Severity Scale	
1	No noticeable effect
2	Minor performance degradation
3	Degraded operation, flight impact
4	Severe loss of propulsion or control
5	Catastrophic Failure

Occurrence Scale	
1	Very unlikely (well-controlled, proven design)
2	Unlikely
3	Possible
4	Likely without mitigation
5	Frequent

Detectability Scale	
1	Easily detected during bench testing/analysis
2	Detectable with inspection/testing
3	Difficult to detect pre-operation
4	Very difficult to detect
5	Not detectable prior to operation

Figure 24: Evaluation Scales for Severity, Occurrence, & Detectability

To better manage risk, a RPN acceptability criteria was developed, to guide design decisions. The criteria is shown and included in the attached Risk Assessment document. It should be noted that for those failure modes who are assigned high RPNs, the most analysis has taken place, to mitigate likelihood of failure; for instance, the heat shield and its associated analysis was implemented to mitigate the likelihood of tank shell failure.

Table 5: RPN Acceptability Criteria

RPN Range	**Acceptability**
> 20	Unacceptable - must be addressed
15 < RPN < 19	Permissible - not ideal

10 < RPN < 14	Permissible with caution
5 < RPN < 9	Permissible
0 < RPN < 4	Acceptable

Given the FMEA, three failure modes stood out as risks, given their high RPNs. These three failure modes were: the spillage of fuel because of tank shell failure, the spillage of fuel due to O-ring or clamping failure, and the leakage or spillage of fuel due to fitting failure. All FMEA assessments are included in the attached risk assessment document.

4.2 Hazard and Risk Assessment (HARA)

For any failure mode that was found to pose a risk to health, safety, or the environment, a hazard and risk assessment (HARA) was undertaken. The failure modes were evaluated for hazard severity and probability, using scales derived from [3], and shown in Figure 25.

HAZARD RISK ASSESSMENT MATRIX					
		HAZARD SEVERITY			
		(1) Catastrophic	(2) Critical	(3) Marginal	(4) Negligible
Hazard Probability	(A) Frequent	1A	2A	3A	4A
	(B) Probable	1B	2B	3B	4B
	(C) Occasional	1C	2C	3C	4C
	(D) Remote	1D	2D	3D	4D
	(E) Improbable	1E	2E	3E	4E

Figure 25: Hazard & Risk Assessment Matrix

This HARA resulted in the summary, found in the attached risk assessment document. The highest assigned hazard risk index (HRI) was 1C, for the potential of shell rupture under unforeseen thermal conditions. Each evaluated HRI is included in the risk evaluation document.

5.0 Summary, Lessons Learned, and Future Work

The following section provides a summary of the project outcomes, key lessons learned throughout the design and manufacturing process, and recommendations for future design iterations. A reflection on technical challenges is included to inform future teams undertaking similar projects.

5.1 Summary

This project involved the design, analysis, and fabrication of a fuel system for the QT1.H hybrid UAV, developed for the University of Victoria's Centre for Aerospace Research. The fuel system was designed to meet strict requirements for mass, packaging, CG stability, slosh damping, and thermal management within the existing airframe.

Key outcomes of the project include the successful thermoforming of HDPE tank shells using CNC machined MDF molds, the design and fabrication of a custom brass fuel fitting assembly to meet tight geometric constraints, and the development of a post-and-plate aluminum mounting structure accommodating all existing aircraft components. Structural and thermal simulations conducted in SolidWorks validated the design under 9G launch loads and confirmed the necessity of an aluminum heat shield, reducing the maximum tank temperature from 249°C to a safe

39°C. All leak, pressure, and CG testing was completed successfully on-site at CfAR, validating the system across all key performance requirements. Specific testing results can be found in the attached test report.

5.2 Lessons Learned

Mold design and material selection proved crucial to the thermoforming process. Initial PETG 3D printed molds deformed under HDPE's thermal load, causing a switch to CNC machined MDF molds which are much more stable under high temperatures. Insufficient draft angle and fillet radii in early iterations caused mold release issues, requiring design iterations. These challenges prove that time investment into mold design significantly reduces iteration time and improves final part quality.

Submitting parts to the machine shop as early as possible was critical to avoiding schedule delays. Discussing drawings and design intent directly with the machinists prior to manufacturing was extremely valuable as this highlighted manufacturability concerns early and prevented issues which would have been discovered after parts were completed. Early and open communication with the machine shop is strongly recommended for future project involving custom machined parts.

Consistent engagement with CfAR throughout the design process was essential, as several key design changes were driven by CfAR feedback during design reviews. Engaging with CfAR early allowed these changes to occur before significant manufacturing effort was invested. Clear client scope allowed the team to remain focused on core deliverables.

5.3 Future Work

While the fuel system successfully meets all client objectives and requirements, a couple areas for design improvement were identified throughout the manufacturing and testing process. The following recommendations are proposed for future design iterations.

The current fuel pump mounting bracket requires a specific bolt assembly sequence due to limited access during installation. Future work should explore alternative bracket geometries or fastener configurations to simplify the assembly process.

Regarding the fuel fitting assembly, there is a specific internal hole within the female brass fitting, which is difficult to fully deburr, posing a risk of O-ring damage upon installation. Future iterations should investigate geometry modifications to improve deburring access or incorporate a chamfer at the critical interface to guide the O-ring safely past any sharp edges.

For the fuel tank assembly, the current O-ring and girder sealing configuration was selected for its proven reliability and potential to act as a mounting surface for slosh mitigation baffles. Future work could investigate replacing this with a custom gasket, potentially simplifying or removing the girder entirely to reduce mass and manufacturing complexity, subject to leak testing validation.

6.0 References

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